LM

LEAD MATRIX
BY ARA DISCOVERIES

CONFIDENTIAL

TECHNICAL SOLUTION DOCUMENT

# ARA Lead Matrix CRM Platform

## Solution, Data, API, Security & Infrastructure Architecture

The complete technical implementation of the platform — solution & logical architecture, the MongoDB data model, the full REST API catalogue, authentication & security design, background processing, integrations, deployment topology, and the strategies for logging, caching, scalability, availability and disaster recovery.

PREPARED FOR
Architects · Developers · DevOps · QA

PREPARED BY
Solution Architecture Office

VERSION
1.0 (Baseline)

DATE OF ISSUE
17 July 2026

CLASSIFICATION
Confidential

COMPANION TO
BRD-2026-001 · FSD-2026-001

# Document Control

Governance and configuration of this Technical Solution Document (TSD).

| Attribute | Detail |
|---|---|
| Document Title | ARA Lead Matrix CRM — Technical Solution Document |
| Reference | ARA-TSD-CRM-2026-001 |
| System | ARA Lead Matrix CRM ("New Ara CRM") |
| Companion Documents | BRD (ARA-BRD-CRM-2026-001) · FSD (ARA-FSD-CRM-2026-001) |
| Version / Status | 1.0 — Baseline — Approved for Review |
| Classification | Confidential |
| Date of Issue | 17 July 2026 |
| Codebase baseline | Backend ≈ 21,250 LOC (Node/Express ESM); React SPA; 24 Mongoose models; 18 route groups. |

## Intended Audience

- **Solution / Software Architects** — architecture, data model, integration design.
- **Developers** — module & component structure, API contracts, coding standards.
- **DevOps Engineers** — deployment, infrastructure, configuration, CI/CD, DR.
- **QA Engineers** — testing strategy, validation, error handling.
- **Enterprise Clients** — technical assurance of the delivered solution.

## Approval & Sign-Off

| Role | Responsibility | Signature | Date |
|---|---|---|---|
| Solution Architect | Architecture accuracy | | |
| Engineering Lead | Implementation fidelity | | |
| DevOps Lead | Infra & deployment accuracy | | |
| Security Officer | Security posture review | | |

> **Basis.** Produced from a full source-level review of the delivered repository (backend services, models, routes, middleware, background jobs, utilities, front-end, build & deploy scripts and environment configuration). It describes the system *as built and in production* as of the issue date.

# Revision History

| Version | Date | Author | Summary |
|---|---|---|---|
| 0.1 | 12 Jul 2026 | Solution Architecture Office | Architecture outline & section structure. |
| 0.6 | 15 Jul 2026 | Solution Architecture Office | Data, API, security & integration architecture drafted from source. |
| 0.9 | 16 Jul 2026 | Solution Architecture Office | Deployment, DR, diagrams, risks & recommendations added. |
| **1.0** | 17 Jul 2026 | Solution Architecture Office | Baseline — full 54-section TSD approved. |

# Table of Contents

ARA Lead Matrix is a multi-tenant MERN-stack CRM: a React single-page front-end, a stateless Node.js/Express REST API, a MongoDB (Azure Cosmos DB for MongoDB vCore) datastore, and an in-process background-job engine that synchronises Meta and Google Ads data and reconciles advertising spend into an immutable billing ledger.

| MERN | 24 | 18 | 3 | 2 |
|---|---|---|---|---|
| ARCHITECTURE PATTERN | COLLECTIONS | API ROUTE GROUPS | EXTERNAL INTEGRATIONS | AUTH REALMS |

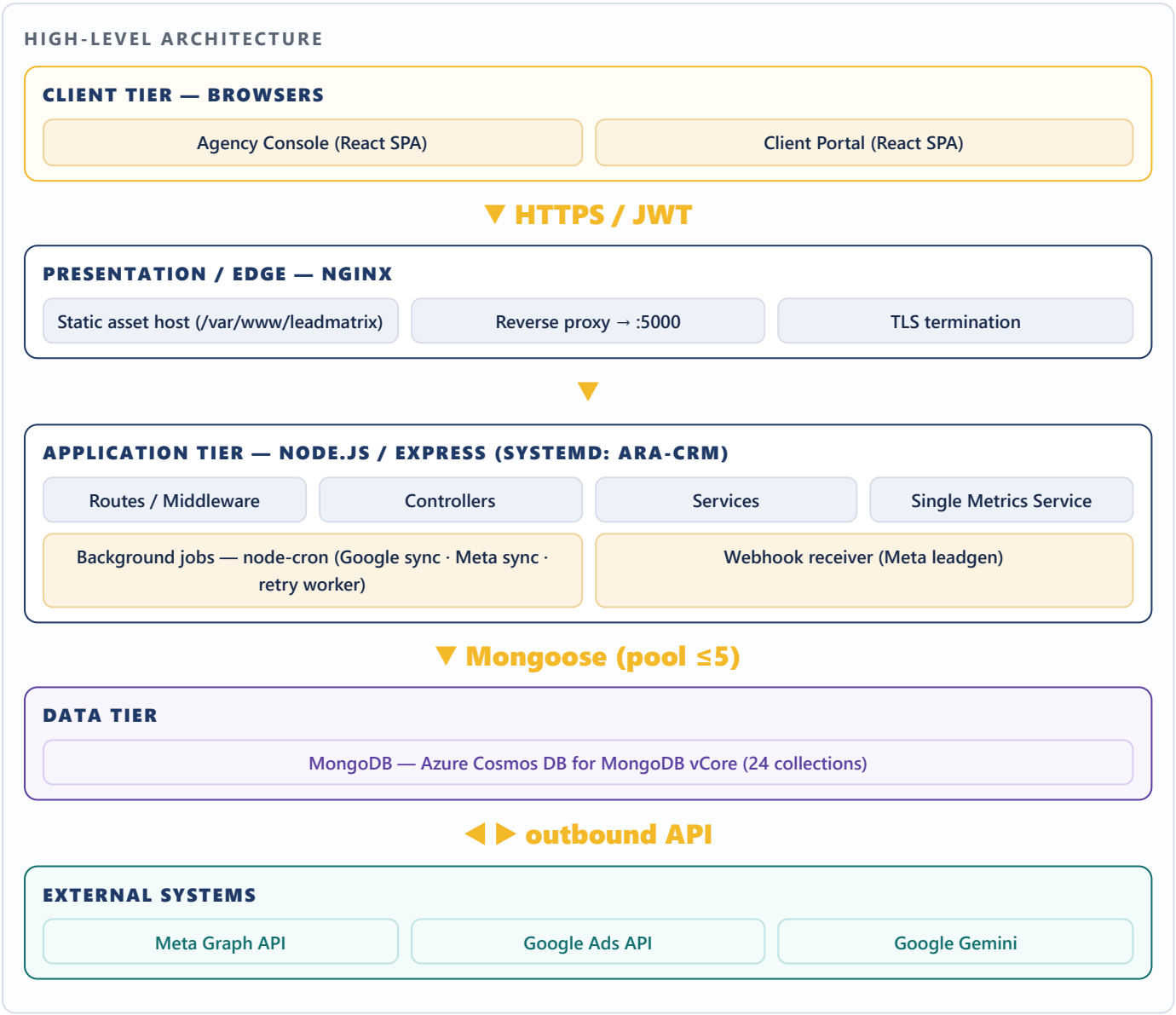
## 5.1 Architectural characteristics

| Characteristic | Implementation |
|---|---|
| Statelessness | JWT-based sessions; no server session store — the API scales horizontally behind a load balancer without sticky sessions (subject to the single-instance scheduler caveat, §33). |
| Multi-tenancy | Row-level tenancy on a shared schema; client identity for portal tokens is derived from the JWT and enforced per request. |
| Idempotency | Unique/compound indexes + idempotency keys guarantee exactly-once lead ingestion and billing writes. |
| Resilience | Typed integration errors, exponential-backoff retries, a webhook dead-letter queue, overlap guards, and boot-time self-healing migrations. |
| Separation of concerns | Routes → Controllers → Services → Models, with a dedicated single metrics service as the canonical KPI authority. |
| Security at rest | bcrypt for passwords; AES-256-CBC for recoverable secrets (page tokens, vault, portal-password recovery). |

## 5.2 Key technical decisions & rationale

| Decision | Rationale |
|---|---|
| Single-database, shared-schema multi-tenancy | Simplicity + cross-client agency reporting; tenancy enforced at the query layer. |
| In-process cron scheduler (node-cron) | Zero extra infrastructure; adequate for current client volume; must run on exactly one instance (§33). |
| Meta insights as reporting source of truth | Avoids under-reporting when lead ingestion is briefly degraded. |
| Delta-based idempotent billing ledger | Financial correctness under retries and concurrency. |
| Post-build JS obfuscation | Raises the bar against trivial client-code inspection (identifier mangling; heavy transforms deliberately disabled for runtime safety). |
| Cosmos DB vCore with small connection pools | Respects a hard per-cluster connection ceiling shared across processes. |

A classic three-tier web architecture augmented with an integration/background-processing tier.




## 6.1 Architectural styles applied

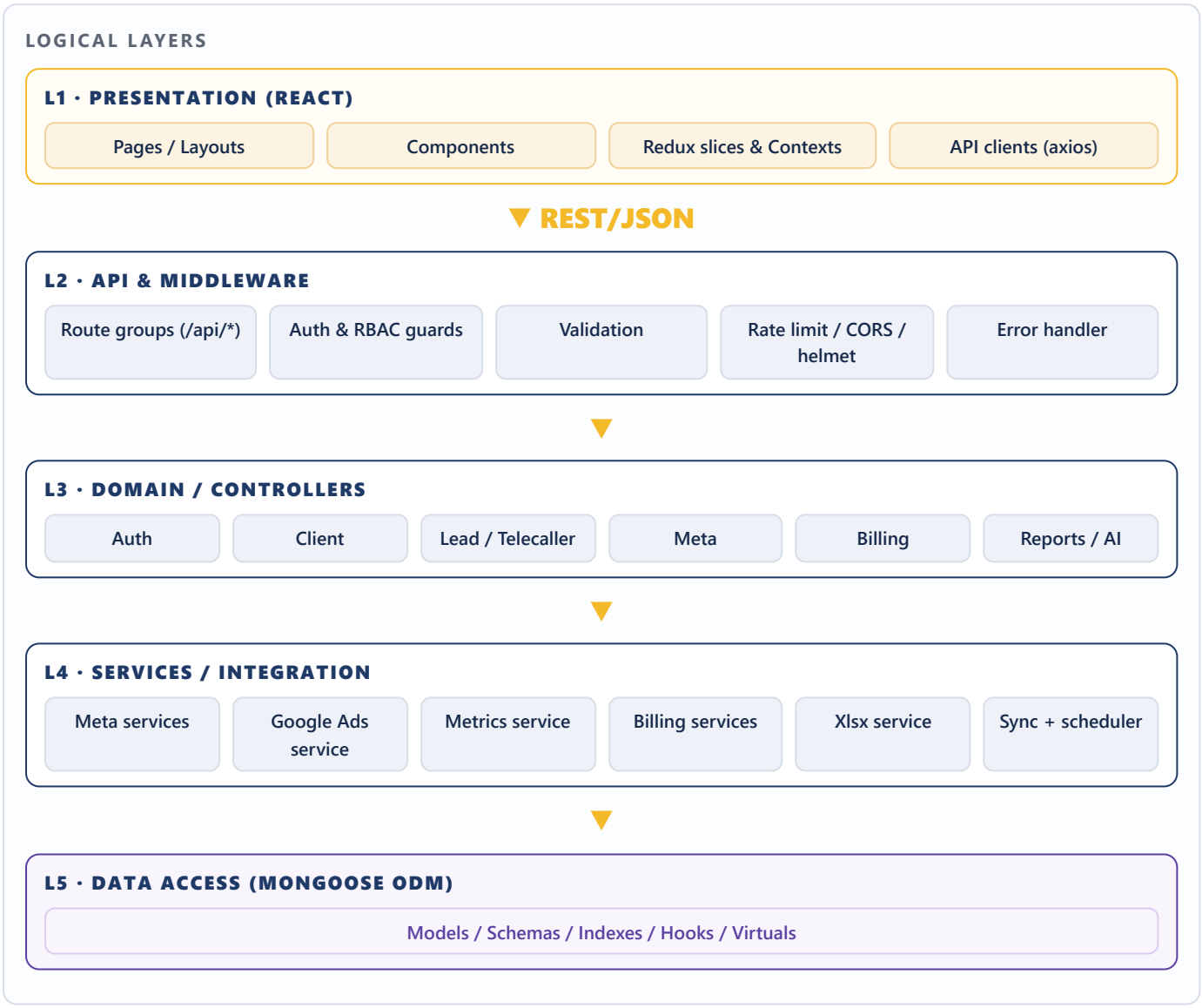
- **Three-tier** (presentation / application / data) with a **REST** API contract.
- **Layered** application internals (routes → controllers → services → models).
- **Event-driven ingestion** for Meta leads (webhook push) with **scheduled batch** reconciliation.
- **Shared-schema multi-tenant SaaS.**

How the business capability is realised technically end-to-end.

| Capability | Technical realisation | Key components |
|---|---|---|
| Identity | Two JWT realms (staff, portal-user); bcrypt credentials; cookie + bearer transport. | authController, clientAuthController, auth middleware, generateToken |
| Ad data acquisition | Scheduled per-client pulls + real-time webhooks; typed errors; rate-limit awareness. | scheduler, syncService, metaSyncService, metaAdsService, googleAdsService |
| Lead ingestion | Signed webhook → raw audit → idempotent upsert → retry queue → poller safety-net. | metaController, metaLeadService, MetaLeadRaw, MetaWebhookRetry |
| Telecalling | 26-column sheet; pre-save automation library; Excel round-trip. | leadTelecallerController, telecallerAutomation, leadXlsxService |
| Billing | Delta reconcile vs snapshot; immutable idempotent ledger; balance cache. | metaBillingService, syncService, BillingTransaction, DailyDebitSnapshot |
| Reporting | One metrics service feeds all surfaces; short-TTL caches. | metaMetricsService, analyticsService, reportsController |
| Intelligence | Gemini REST call with strict-JSON contract + repair. | aiController |
| Presentation | Lazy-loaded React SPA; Redux slices + context caches; MUI. | App.js, store, contexts, pages, components |

## 7.1 Cross-cutting concerns

| Concern | Mechanism |
|---|---|
| Security | helmet, CORS allow-list, two-tier rate limiting, JWT verify, tenant scoping, encryption utilities. |
| Consistency | Canonical metrics service; shared enum constants (front/back in lockstep). |
| Observability | MetaSyncRun log, retry queue, raw-lead audit, morgan HTTP logs, tagged diagnostics. |
| Data integrity | Unique/compound/partial indexes; idempotency keys; boot-time self-heal. |

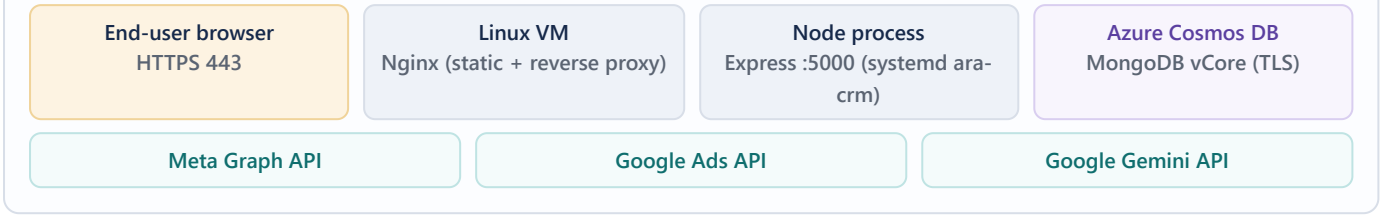



Each layer depends only on the layer beneath it. Controllers never talk to external APIs directly — they delegate to services; services own all Graph/Ads API traffic and normalisation.

## 8.1 Front-end logical state

| Store / context | Responsibility |
|---|---|
| auth slice | Staff session, role, permissions; persisted to localStorage. |
| clients / leads / dailyEntry / dailyLeadData slices | Entity lists, CRUD, stats. |
| DataCacheContext | Caches slow-changing client list across pages. |
| ThemeContext | Brand colours / dark mode. |

# Physical Architecture




| Node | Detail |
|---|---|
| Web/edge | Nginx serves the built SPA from `/var/www/leadmatrix` and reverse-proxies `/api` to the Node backend; TLS termination. Public host: `leadmatrix.discovertechnologies.co` (also `crm.aradiscoveries.com` allow-listed). |
| Application | Single Node.js process managed by `systemd` unit `ara-crm`, listening on port 5000. Hosts the API and the in-process cron scheduler. (A managed Azure App Service endpoint also exists as an alternate/observed target.) |
| Database | Azure Cosmos DB for MongoDB vCore — TLS, SCRAM-SHA-256 auth, `retrywrites=false`, shared per-cluster connection ceiling (pool sized small, default 5/process). |
| External | Outbound HTTPS to Meta, Google Ads and Gemini; inbound HTTPS webhook from Meta to `/api/meta/webhook`. |

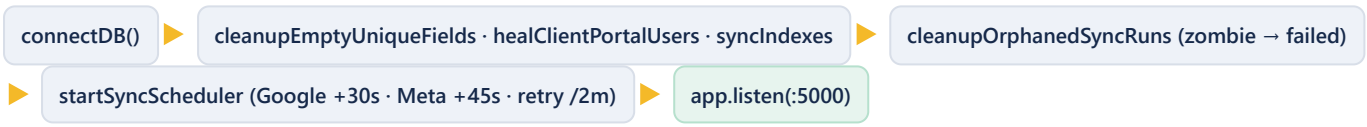
> **Note.** The frontend base URL is environment-driven ( `REACT_APP_API_URL` ), currently `https://leadmatrix.discovertechnologies.co/api`. The codebase also carries an Azure App Service fallback URL, indicating the app has run on both a self-managed VM and Azure App Service.

## 10.1 Deployment procedure (redeploy.sh)

```
1. git pull                    # ~/ara-crm — fetch latest code
2. sudo systemctl restart ara-crm  # restart Node backend (systemd)
3. cd crm-frontend && npm run build # CRA build + post-build obfuscation
4. sudo cp -r build/* /var/www/leadmatrix/    # publish static SPA
5. sudo chown -R www-data:www-data /var/www/leadmatrix
   ✅ Deploy done
```

| Aspect | Detail |
|---|---|
| Trigger | Manual (operator runs `redeploy.sh`); `set -e` aborts on first failure. |
| Backend release | In-place git pull + systemd restart (no blue-green; brief restart gap). |
| Frontend build | `react-scripts build` then `scripts/obfuscate.js` (javascript-obfuscator, light preset — identifier mangling; heavy transforms disabled for React 19/MUI/Recharts safety). |
| Frontend release | Static files copied to the Nginx web root; ownership handed to `www-data`. |
| Boot behaviour | On start: connect DB → self-heal migrations → orphaned-sync cleanup → start scheduler → listen. Fatal boot error exits the process (systemd restarts). |

## 10.2 Startup sequence



| Component | Technology | Role |
|---|---|---|
| Compute | Linux VM (systemd) | Runs the Node.js backend as service `ara-crm`. |
| Web server | Nginx | Static hosting of the SPA + reverse proxy + TLS. |
| Runtime | Node.js (ESM) | Express API + in-process scheduler. |
| Database | Azure Cosmos DB for MongoDB vCore | Managed, replicated document store. |
| DNS / TLS | Public domains + HTTPS | `leadmatrix.discovertechnologies.co`, `crm.aradiscoveries.com`. |
| Process supervision | systemd | Auto-restart on crash/exit. |
| Secrets | Environment files (`.env`) | DB URI, JWT/encryption keys, API tokens. |

> **Observation.** The infrastructure is a single-VM deployment (no container orchestration, no autoscaling group, no managed secret vault). This is appropriate for the current scale but is the primary constraint for the availability and scalability strategies (§33–§36).

## 12.1 Backend

| Concern | Library |
|---|---|
| Runtime / framework | Node.js (ESM) · Express 5 |
| ODM / database | Mongoose 9 · MongoDB (Cosmos vCore) |
| Auth / crypto | jsonwebtoken · bcryptjs · Node crypto (AES-256-CBC) |
| Security middleware | helmet · cors · express-rate-limit · cookie-parser |
| Validation | express-validator |
| Scheduling | node-cron |
| Integrations | google-ads-api · googleapis · Meta Graph (via fetch) · Gemini (REST) |
| Files / logging | exceljs · multer · morgan |
| Dev | nodemon |

## 12.2 Frontend

| Concern | Library |
|---|---|
| Framework | React 19 · react-router-dom 6 |
| State | Redux Toolkit · react-redux · React Context |
| UI | MUI (Material + Joy + Base + System) · Emotion |
| Data viz | @mui/x-charts · Recharts |
| Grids / pickers | @mui/x-data-grid · @mui/x-date-pickers |
| HTTP / forms | axios · Formik · Yup · date-fns |
| Export | xlsx · xlsx-js-style · jspdf · jspdf-autotable · html2canvas · file-saver |
| Build / harden | react-scripts (CRA) · javascript-obfuscator |

```
New_Ara_CRM/
├── backend/                # Node.js / Express API + jobs
│   ├── config/db.js        # Mongo connection + boot self-heal
│   ├── controllers/        # 12 controllers (request handlers)
│   ├── routes/             # 18 route groups (/api/*)
│   ├── middleware/         # auth, permissions, validation, errorHandler
│   ├── models/             # 24 Mongoose schemas
│   ├── services/           # Meta/Google/metrics/billing/xlsx/AI
│   ├── sync/               # scheduler, syncService, metaSyncService
│   ├── lib/                # telecallerAutomation
│   ├── utils/              # encryption, tokens, mappers, rate-limit, signature
│   ├── scripts/            # migrations, smoke tests, backfills
│   ├── seedAdmin.js        # seed a first admin
│   ├── server.js           # app bootstrap + middleware wiring
│   └── .env(.example)      # backend configuration
├── crm-frontend/           # React SPA
│   ├── public/
│   ├── scripts/obfuscate.js # post-build hardening
│   └── src/
│       ├── api/            # axios instances + resource clients
│       ├── pages/          # 23 routed screens
│       ├── components/     # dashboard/, telecaller/, shared widgets
│       ├── layouts/        # MainLayout (agency shell)
│       ├── store/slices/   # Redux Toolkit slices
│       ├── contexts/       # DataCache, Theme
│       ├── constants/      # telecallerSheet (enum lockstep)
│       └── utils/          # ProtectedRoute, exporters
├── docs/                   # BRD, FSD, TSD, integration references
└── redeploy.sh             # manual deployment script
```

| Folder | Responsibility |
|---|---|
| backend/config | DB connection, pool sizing, boot-time self-heal & index repair. |
| backend/routes | HTTP surface; binds middleware guards to controller actions. |
| backend/controllers | Request/response orchestration & business logic per resource. |
| backend/middleware | auth (protect/protectClient/protectAdminOrClient/authorize), permissions (checkPermission/checkRole + maps), validation, errorHandler. |
| backend/models | Schemas, indexes, virtuals, hooks — the data contract. |
| backend/services | External API clients, normalisation, metrics, billing, Excel, AI. |
| backend/sync | Cron scheduler + Google/Meta sync orchestration. |
| backend/lib | Framework-free domain logic (telecaller automation). |
| backend/utils | Crypto, JWT, Meta field mapping/rate-limit/signature, validators. |
| backend/scripts | One-off migrations, backfills, and operational smoke tests. |
| crm-frontend/src/api | Axios base (interceptors, 401 grace) + per-resource clients. |
| crm-frontend/src/pages | Route-level screens (agency + portal). |
| crm-frontend/src/components | Reusable UI (dashboard widgets, telecaller grid, reports). |
| crm-frontend/src/store | Redux Toolkit store & slices. |

Backend modules map route group → controller → service(s) → model(s).

| Module | Route | Controller / Service | Primary models |
|---|---|---|---|
| Auth | /auth, /client-portal | authController, clientAuthController | User, ClientPortalUser, Client |
| Users | /users | userController | User |
| Clients | /clients | clientController | Client (+cascade) |
| Leads | /leads | leadController + metaMetricsService | Lead |
| Meta / Telecalling | /meta | metaController, leadTelecallerController, metaLeadService, metaAdsService, metaBillingService, telecallerAutomation, leadXlsxService | Meta*, Lead, BillingTransaction |
| Google Ads | /google-ads, /metrics, /analytics | googleAdsService, syncService, analyticsService | Campaign, Metric, Keyword |
| Daily data | /daily-entries, /daily-lead-data | dailyEntryController, dailyLeadDataController | DailyEntry, DailyLeadData |
| Billing | /payments, /billing | payments/billing routes + metaBillingService | Payment, BillingTransaction, DailyDebitSnapshot |
| Reports | /reports | reportsController | various (aggregations) |
| Content | /content-entries | contentEntryController | ContentEntry |
| Vaults | /vault, /personal-vault | route handlers + encryption util | Vault, PersonalVault |
| AI | /ai | aiController | — (stateless) |

## 16.1 Backend components

| Component | Responsibility |
|---|---|
| Metrics service | Single authority for lead/spend KPIs (form vs WhatsApp rules, CPL/CTR/CPC/CPM, ingestion gap). |
| Meta ad service | Graph API client: retries, pagination, rate-limit parsing, field sets. |
| Meta lead service | Idempotent ingest, retry queue processing, poller. |
| Billing services | Delta reconcile, idempotent ledger writes, balance cache. |
| Telecaller automation | Framework-free pre-save rules (reminder/history/DARMANT/stamps). |
| Xlsx service | Import parse + value cleaning + upsert; export 26-column workbook. |
| Scheduler | Cron registration, overlap guards, run-state, retry worker. |

## 16.2 Frontend components

| Component group | Members |
|---|---|
| Shell | MainLayout, ProtectedRoute, OfflineScreen, LeadMatrixLoader. |
| Dashboard | BalanceWatch, PerformanceInsights, KpiStrip, PlatformOverview, ClientOverviewCards, SideWidgets, ActiveCampaigns. |
| Telecalling | TelecallerSheet, QueueChips, TelecallerFilterBar, ImportLeadsDialog, MetaLeadsTable. |
| Reporting | TelecallingReport, MonthlyAbstract, MetricsBand, PortalDashboardToday, AiCampaignInsights. |
| Alerts | LowBalanceBell. |

| Layer | Contains | Rules |
|---|---|---|
| Routing | Express routers | Only wire middleware + controller; no business logic. |
| Middleware (cross-cut) | auth, permissions, validation, rate-limit, error handler | Guard, validate, normalise; short-circuit on failure. |
| Controller (domain) | Per-resource handlers | Orchestrate; may call services + models; own transaction-like flows. |
| Service (integration/domain) | External clients, metrics, billing, xlsx, automation | All external I/O and heavy computation; reusable across controllers. |
| Data access (ODM) | Mongoose models | Schema, indexes, hooks, virtuals; the only DB touch-point. |

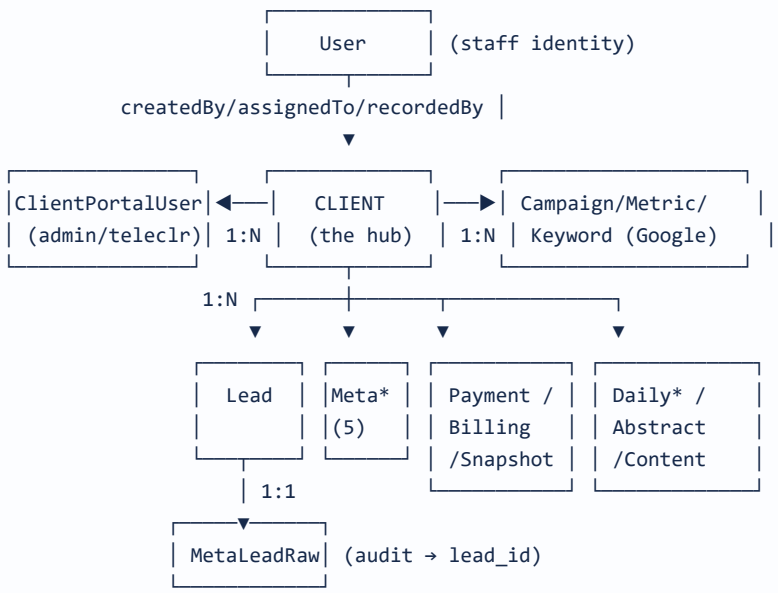
**Dependency rule.** Dependencies point downward only. Controllers depend on services and models; services depend on models and utils; models depend on nothing above them. The single metrics service is shared by multiple controllers to prevent KPI drift.

| Aspect | Detail |
|---|---|
| Engine | MongoDB API on Azure Cosmos DB for MongoDB vCore (document model). |
| ODM | Mongoose 9 — schemas, validation, hooks, virtuals, index sync. |
| Connection | TLS + SCRAM-SHA-256; `retrywrites=false`; pool `maxPoolSize` default 5 (env `DB_MAX_POOL`), `minPoolSize` 0, idle 120 s. |
| Indexing | `autoIndex` enabled in non-prod only; prod relies on explicit `syncIndexes()`; unique/compound/partial indexes enforce integrity & idempotency. |
| Multi-tenancy | Row-level via `client` / `client_id` foreign keys on shared collections. |
| Integrity patterns | Idempotency keys, partial unique indexes (non-empty), compound daily keys `{client, date, …}`, immutable audit/ledger collections. |
| Self-heal | Boot-time cleanup of empty unique fields, stale portal-user indexes, and orphaned sync runs. |

## 18.1 Collection groups

| Group | Collections |
|---|---|
| Identity | User, ClientPortalUser |
| Core | Client, Lead |
| Google Ads | Campaign, Metric, Keyword |
| Meta Ads | MetaCampaign, MetaAdSet, MetaAd, MetaInsights, MetaLeadForm, MetaLeadRaw, MetaWebhookRetry, MetaSyncRun |
| Finance | Payment, BillingTransaction, DailyDebitSnapshot |
| Operational | DailyEntry, DailyLeadData, AbstractEntry, ContentEntry |
| Secrets | Vault, PersonalVault |

CLIENT-CENTRED ER (HUB-AND-SPOKE)




Client is the operational hub (parent of ~15 collections); User is the identity hub. The Meta ad hierarchy (MetaCampaign → MetaAdSet → MetaAd → MetaInsights) is linked by business string IDs, each also referencing Client. MetaLeadRaw links 1:1 to the created Lead via `lead_id`.

## 19.1 Cardinality summary

| Relationship | Cardinality |
|---|---|
| Client → ClientPortalUser | 1 : N |
| Client → Lead / Meta* / Google* / Finance / Daily* | 1 : N (each) |
| User → Lead (assignedTo, createdBy) | 1 : N |
| User ↔ PersonalVault.sharedWith | M : N |
| Payment → BillingTransaction (reference.payment_id) | 1 : 1 |
| MetaLeadRaw → Lead (lead_id) | 1 : 1 |
| ClientPortalUser → ClientPortalUser (createdBy) | self-ref |

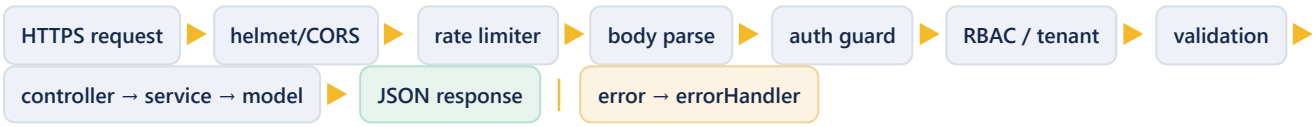
| Collection | Key fields | Notable indexes / constraints |
|---|---|---|
| User | userID(uniq), email(uniq), password, role, permissions[], isActive | unique email/userID; role, isActive indexes; bcrypt hook; USRnnn generator. |
| Client | clientName, status, google_ads_customer_id, meta_ad_account_id, billing{}, meta_pages[], telecalling_targets{}, portal* | unique+sparse on ad IDs; text index; validators (10-digit / act_). |
| ClientPortalUser | clientId, username, email, password, role, isActive | unique {clientId,username}; partial unique {clientId,email}. |
| Lead | name, email, contact, source*, status*, meta_*, follow_ups[], telecaller cols | unique {client,contact} partial; unique sparse meta_leadgen_id; many query indexes; virtuals. |
| Campaign / Metric / Keyword | client_id, campaign_id, metrics/KPIs, date | unique campaign_id; compound {client,campaign,date}; unique keyword {client,campaign,criterion,date}. |
| MetaCampaign/AdSet/Ad | client_id, *_id, status, budgets, targeting | unique on entity id; {client,status}/{client,campaign} indexes. |
| MetaInsights | client_id, level, entity_id, date, spend, leads, KPIs | unique {client,entity,date}; {date,level} cross-client. |
| MetaLeadForm | form_id, page_id, client_id(nullable), question_schema[] | unique form_id; nullable client for unmapped forms. |
| MetaLeadRaw | leadgen_id(uniq), raw_payload, processed, lead_id | unique leadgen_id; {processed, received_at}. |
| MetaWebhookRetry | leadgen_id, payload, attempts, next_attempt_at, status | {status, next_attempt_at} worker query. |
| MetaSyncRun | run_id(uniq), scope, status, counts{}, errors[] | unique run_id; {status, started_at}. |
| Payment | client_id, amount, method, date | client_id, date indexes. |
| BillingTransaction | client_id, type, amount, balance_after, source, idempotency_key | {client,occurred_at}; unique sparse idempotency_key. |
| DailyDebitSnapshot | client_id, platform, campaign_id, date, debited/reported | unique {client,platform,campaign,date}. |
| DailyEntry / DailyLeadData | date, client, meta/google counts, CPL | unique {date,client} / unique date; pre-save calc. |
| AbstractEntry | client, date(YYYY-MM-DD), manualValues(Map) | unique {client,date}. |
| ContentEntry | date, clientName, contentType, status, approvalStatus | date/clientName/status indexes. |
| Vault / PersonalVault | credentials (AES), platform, sharedWith[] | encrypted password at rest. |

| Type | Relationships |
|---|---|
| ObjectId references (enforced by app) | Lead.client→Client, Lead.assignedTo/createdBy→User, ClientPortalUser.clientId→Client, Payment.client_id→Client, BillingTransaction.client_id→Client & reference.payment_id→Payment, MetaLeadRaw.lead_id→Lead, all Google/Meta facts.client_id→Client, PersonalVault.createdBy/sharedWith→User, AbstractEntry.client→Client, MetaSyncRun.client_id→Client. |
| String / business-key links | Meta hierarchy cross-links (campaign_id/adset_id/ad_id); Vault.clientId (string); DailyEntry.clientId (string); ContentEntry.clientName (string). |
| Non-client-scoped | DailyLeadData is a global per-day aggregate (unique on date). |
| Financial flow | Payment (credit) → BillingTransaction → Client.billing.available_balance; ad spend (Metric/MetaInsights) → delta → DailyDebitSnapshot → BillingTransaction (debit). |

**Referential integrity** is application-enforced (document DB). Hard-delete of a client cascades explicitly across ~10 collections with per-collection outcome reporting; drop is a reversible soft-delete preserving all references.

| Aspect | Detail |
|---|---|
| Style | RESTful JSON over HTTPS, mounted under `/api`. |
| Envelope | Success `{ success:true, data\|message\|… }`; error `{ success:false, error\|message\|errors }`. |
| Auth transport | `Authorization: Bearer` or httpOnly `token` cookie ( `Secure; SameSite=None` ). |
| Guards | protect · protectClient · protectAdminOrClient · authorize(roles) · requireClientPortalRole · checkPermission · checkRole · requireAgencyToken. |
| Rate limiting | Auth 10/15 min (reset on success); API 1000/15 min per IP. |
| Body limits | 10 MB JSON/urlencoded; raw body captured only for the Meta webhook (HMAC). |
| Caching | Short-TTL in-memory caches on hot read endpoints (analytics 5 min; lead pivots 60 s; user cache 1 min). |
| Errors | Centralised handler maps Mongoose/JWT errors to status + message. |

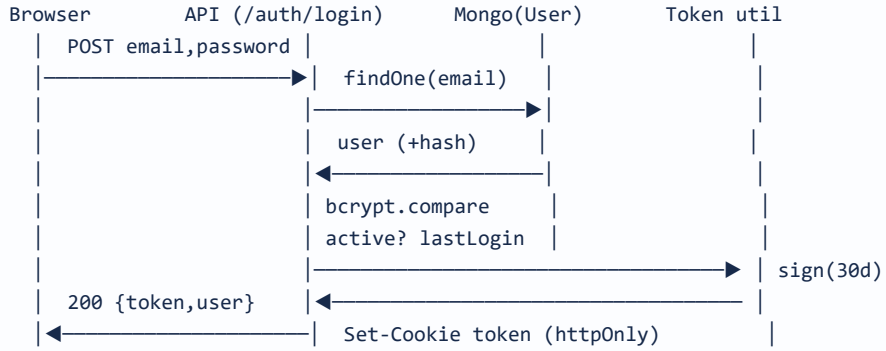
## 22.1 Request lifecycle



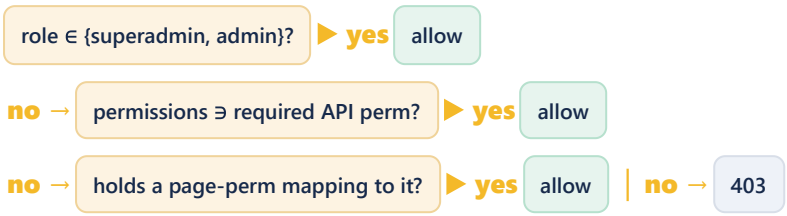
| Group | Guard | Representative endpoints |
|---|---|---|
| `/auth` | public/protect | POST login · POST client-login · POST register · GET me · GET client-me · POST logout · PUT update-details · PUT update-password |
| `/users` | admin/superadmin | GET / · POST / · GET stats · GET roles · GET teams · GET/PUT/DELETE :id · PATCH toggle-status · PATCH change-password · PATCH permissions |
| `/clients` | checkPermission | GET / · POST / · GET stats · GET/PUT/DELETE :id · PATCH status · PATCH drop · PATCH reonboard · GET :id/portal-credentials · GET smm-users |
| `/leads` | checkPermission | GET / · POST / · GET stats/summary · GET monthly-meta-by-client · GET daily-by-client · GET/PUT/DELETE :id |
| `/meta` | protectAdminOrClient | GET/POST webhook · sync(+historical) · unassigned-forms · forms/:id/assign\|reprocess · client/:id/config\|analytics\|spend-overview · client/:id/leads (GET/POST/PUT/DELETE) · leads/import\|export · telecalling-report · telecalling-targets · monthly-abstract(+/cell) · sync-runs · retry-queue · raw-leads |
| `/google-ads` | admin/superadmin | POST sync · sync/:id · GET campaigns/:id · PUT customer-id\|enable\|associate · POST clients/bulk-associate · GET clients/unassociated · POST validate-customer-id · GET sync-status |
| `/metrics` | protectAdminOrClient | GET dashboard/:id · campaigns/:id · daily/:id |
| `/analytics` | protectAdminOrClient | GET clients · client/:id · client/:id/spend-overview |
| `/daily-entries` | checkPermission | CRUD · stats/summary\|today · date/:date · meta-lead\|meta-fund · sync-meta · main-* proxies |
| `/daily-lead-data` | checkPermission | CRUD · stats/summary · stats/campaign-comparison · date/:date |
| `/payments` | admin/superadmin | POST / · GET :clientId |
| `/billing` | admin/superadmin | POST :id/reconcile · :id/deep-sync · :id/reset(confirm) |
| `/client-portal` | protectClient | GET analytics · GET/POST/PUT/DELETE users (admin) |
| `/reports` | report:view | GET dashboard · sales · campaign-performance · lead-performance · user-performance* · financial* (*admin) |
| `/content-entries` | checkPermission | CRUD · calendar/:year/:month · smm-users |
| `/vault · /personal-vault` | protect / admin | credential CRUD (decrypt on read); personal share PATCH |
| `/ai` | protect | POST analyze-campaign |
| `/health · /` | public | health check & root banner |

| Aspect | Detail |
|---|---|
| Mechanism | Stateless JWT (jsonwebtoken), signed with `JWT_SECRET`. |
| Staff token | Payload `{ id }`; login by email+password; access 30 d, refresh 90 d (`JWT_REFRESH_SECRET`). |
| Portal token | Payload `{ id, role, clientId, userId }`; login by username/email+password, client-scoped. |
| Transport | Bearer header or httpOnly cookie `token` (`Secure; SameSite=None` for cross-site HTTPS). |
| Password storage | bcrypt (salt rounds 10) in pre-save hooks; `select:false`. |
| Guards | `protect` (staff; 1-min user cache; active check), `protectClient` (portal; portalEnabled + active), `protectAdminOrClient` (dual). |
| Legacy self-heal | Portal login falls back to legacy client credentials and auto-seeds an admin ClientPortalUser; tokens without `userId` degrade to least-privilege on protected portal routes. |
| Client 401 grace | Front-end delays forced logout 800 ms and cancels it if any success lands — resists transient 401s. |

## 24.1 Login sequence



## 25.1 Agency RBAC — evaluation order




Page↔API map (e.g. page `daily-entry` / `daily-lead-data` grants `entry:create`; page `clients` / `dashboard` grants `client:read`). Roles are free-form; only superadmin/admin are hard-coded full access.

## 25.2 Portal RBAC & tenancy

| Control | Behaviour |
|---|---|
| Role gate | `requireClientPortalRole('admin')` on user-management & ad/billing routes; telecaller limited to leads/reports. |
| Tenant isolation | `clientId` taken from the JWT; shared routes reject mismatched `:clientId` (403 "Portal token cannot access another client"). |
| Last-admin guard | Cannot demote/deactivate/delete the final active admin; cannot delete self. |

> **Gap.** `/api/vault` has authentication but no role/permission gate and returns decrypted client credentials to any authenticated staff user (see §26, §50, §53).

| Control | Implementation |
|---|---|
| Transport security | HTTPS end-to-end; secure cookies; HSTS via edge. |
| HTTP hardening | helmet (cross-origin resource policy set for asset sharing). |
| CORS | Credentialed, explicit origin allow-list (prod domains + localhost). |
| Rate limiting | Strict login limiter + generous API limiter. |
| Password security | bcrypt (rounds 10); never returned; `select:false`. |
| Secret encryption | AES-256-CBC (random IV, `iv:ciphertext`) for page tokens, vault, recoverable portal password; SHA-256 hashing helper. |
| Webhook integrity | HMAC-SHA256 over raw body vs `X-Hub-Signature-256`; verify-token challenge. |
| Injection resistance | Mongoose parameterised queries; express-validator input validation; enum allow-lists. |
| Tenant isolation | Token-derived client identity; per-route scoping. |
| Client hardening | Production bundle obfuscation (identifier mangling). |

## 26.1 Threats & mitigations

| Threat | Mitigation |
|---|---|
| Credential brute force | Auth rate limiter (10/15 min). |
| Cross-tenant access (IDOR) | JWT-derived tenant + per-route assert. |
| Token theft | httpOnly secure cookies; short-lived intent (30 d) — see §53 for rotation. |
| Webhook spoofing | HMAC signature verification. |
| Secret exposure | Encryption at rest; keys in env (see §53 for a vault recommendation). |

| Log | Detail |
|---|---|
| HTTP access | morgan — `combined` in production, `dev` otherwise. |
| Application logs | Structured, tagged console logs ( `[sync-scheduler]` , `[meta-scheduler]` , `[meta-retry]` , `[client-login]` , `[server]` ) captured by systemd/journald. |
| Sync audit | MetaSyncRun documents per run (counts, per-stage errors, rate-limit usage). |
| Lead audit | MetaLeadRaw immutable payload log; MetaWebhookRetry failure trail. |
| Financial audit | Append-only BillingTransaction ledger with idempotency keys. |
| Sensitive data | Login diagnostics never log passwords; errors omit stack traces in production. |

> **Recommendation.** Introduce centralised, structured JSON logging (e.g. pino) shipped to an aggregator for search, retention and alerting (§53).

| Signal | Source |
|---|---|
| Liveness | `GET /health` (uptime, timestamp) and `GET /` banner. |
| Sync health | `GET /api/meta/sync-status`, `/sync-runs`, `/google-ads/sync-status` (last run, duration, status, counts, errors). |
| Retry backlog | `GET /api/meta/retry-queue` (pending/abandoned). |
| Ingestion gap | Lead pivots expose `missingFormLeads` (Meta-reported minus CRM-held). |
| Balance health | Low-balance bell + Balance Watch as operational monitors. |
| Process | systemd status / journald for crashes & restarts. |

> **Recommendation.** Add uptime checks on `/health`, alerting on failed/partial sync runs and growing retry backlog, and DB/host metrics (§53).

| Condition | Status | Message |
|---|---|---|
| CastError (bad ObjectId) | 404 | "Resource not found" |
| Duplicate key (11000) | 400 | "<field> already exists" |
| ValidationError | 400 | joined field messages |
| JsonWebTokenError | 401 | "Invalid token" |
| TokenExpiredError | 401 | "Token expired" |
| Permission/role denial | 403 | "Access denied. Required permission/role: ..." |
| Tenant violation | 403 | "Portal token cannot access another client" |
| Unknown route | 404 | "Not Found - <url>" |
| Rate limit | 429 | "Too many ..., please try again later." |
| Fallback | 500 | "Server Error" (stack in dev only) |

All async handlers are wrapped ( `asyncHandler` ) so rejections route to the central handler. Process-level `unhandledRejection` / `uncaughtException` handlers log and exit for a clean systemd restart.

**Consistency note.** Three error-payload key conventions coexist ( `message` / `error` / `errors` ) and duplicate conflicts return 409 (portal users) vs 400 (staff users); 422 is unused. Standardisation recommended (§53).

THREE-LAYER VALIDATION

L1 Client (Formik/Yup + inline regex) — instant feedback ▶ L2 API (express-validator + controller enum checks) — 400 {errors[]} ▶ L3 Schema (Mongoose validators/enums/indexes) — final guard

| Rule class | Examples |
|---|---|
| Format | email regex; Google ID 10 digits (schema) / 8–12 (UI); Meta ID `act_\d+` ; username `^[a-z0-9._-]{3,60}$` . |
| Range/length | password ≥ 6; client name 2–200; notes ≤ 1000; pagination limit ≤ 10000; date range ≤ 90 days. |
| Enum allow-lists | status, source, telecalling vocabulary (validated by controller before save). |
| Uniqueness | email, userID, ad IDs (sparse), contact per client (partial), leadgen_id, idempotency key. |
| Business guards | billing fields stripped from client update; last-admin protection; confirm-required destructive ops. |

| Technique | Detail |
|---|---|
| Targeted indexing | Query-shaped indexes on hot paths (client+date, meta_created_time, status, form/campaign/ad ids). |
| Bulk endpoints | dashboard-overview & daily-metrics replace N+1 per-client calls. |
| Parallelism | Analytics computed in parallelised Promise.all batches; webhook events processed via Promise.allSettled. |
| Short-TTL caching | Analytics 5 min, lead pivots 60 s, user lookups 1 min. |
| Aggregation at DB | Roll-ups computed via Mongo aggregation, not in-app loops. |
| Field projection | Meta field sets avoid `fields=*`; select only what's needed. |
| Front-end | Route-level code splitting + prefetch; silent background refresh; memoised computed views; balance cache for fast reads. |
| Sync hygiene | Sequential per-client sync, staggered cadence, overlap guards to avoid contention. |

| Cache | Scope / TTL | Purpose |
|---|---|---|
| Analytics cache | In-process · 5 min | Expensive per-client Google analytics. |
| Lead-pivot cache | In-process · 60 s | monthly-meta-by-client pivots. |
| User cache | In-process · 1 min | Auth middleware user lookups. |
| Balance cache | Client.meta_ad_account_balance | Fast dashboard balance reads (scraped during sync). |
| Front-end caches | DataCacheContext + 5-min client-ad cache | Avoid re-fetching slow-changing data across pages. |
| Main-API proxy cache | 5 min | Companion-service clients/leads/funds lookups. |

> **Note.** Caches are per-process and in-memory — correct for a single instance. Horizontal scaling would require a shared cache (Redis) to stay consistent (§33).

| Dimension | Current & path forward |
|---|---|
| API tier | Stateless JWT enables horizontal scale-out behind a load balancer. Caveat: in-memory caches and the in-process scheduler assume a single instance. |
| Scheduler | Must run on exactly one node. To scale the API, extract sync/webhook workers into a dedicated job process (or leader-elect) so only one runs cron. |
| Database | Cosmos vCore scales vertically/with shards; connection pools kept small due to a hard per-cluster ceiling — more app instances need pool budgeting. |
| Caching | Move to Redis for a shared cache across instances. |
| Rate limiting | express-rate-limit is per-process/in-memory; a shared store (Redis) needed for multi-instance correctness. |
| Vertical growth | Sequential per-client sync scales linearly with client count; parallelise with bounded concurrency + queue as the roster grows. |

# 34 High Availability Strategy

| Layer | Current posture | Recommended |
|---|---|---|
| Application | Single VM + systemd auto-restart on crash. | ≥2 instances behind a load balancer; health-based routing. |
| Database | Managed Cosmos DB (built-in replication/HA). | Confirm zone-redundant tier + failover SLA. |
| Web/edge | Single Nginx. | Redundant edge / managed CDN for static SPA. |
| Jobs | In-process scheduler (single point). | Dedicated worker with restart + leader election. |
| Resilience built-in | Overlap guards, zombie-run cleanup, retry queue, poller safety-net, graceful degradation. | — |

**Current SPOF.** The single application VM is the main availability risk; the database tier is managed and already highly available.

| Asset | Approach |
|---|---|
| Database | Azure Cosmos DB automatic backups (platform-managed, point-in-time depending on tier). **Action:** confirm retention window & restore runbook. |
| Application code | Git repository is the source of truth; deploy is a git pull. |
| Configuration/secrets | `.env` files on the host — **not** in version control; must be backed up securely out-of-band. |
| Immutable audit | Ledger + raw-lead + sync-run collections provide replay/audit even without file backups. |
| User-generated exports | Excel/PDF generated on demand — no separate backup needed. |

> **Recommendation.** Document and test DB restore; store `.env` secrets in a managed vault with backup; schedule periodic logical exports for an independent recovery copy (§53).

| Scenario | Recovery approach |
|---|---|
| Application host loss | Re-provision VM, install Node/Nginx, restore `.env`, `git pull`, `redeploy.sh`. Boot self-heal repairs indexes; scheduler resumes; retry queue & poller recover missed leads. |
| Database loss/corruption | Restore from Cosmos backup to last good point; the ledger/raw-lead audit enables reconciliation of any gap. |
| Integration outage | Sync skips/backs off; leads queue in retry; balances resume on recovery — no data loss. |
| Secret compromise | Rotate JWT/encryption keys & API tokens; re-encrypt secrets; force re-login. |

| Objective | Indicative target (to formalise) |
|---|---|
| RTO (app) | Hours — bounded by VM re-provision + deploy. |
| RPO (data) | Minutes-to-hours — bounded by Cosmos backup granularity; near-zero for leads/billing thanks to idempotent replay. |

> **Recommendation.** Formalise RTO/RPO, automate infrastructure provisioning (IaC), and rehearse a restore drill (§53).

| Stage | Current state |
|---|---|
| Source control | Git (GitHub) — single `main` branch workflow observed. |
| Continuous integration | **None** — no pipeline ( `.github/workflows` absent); no automated build/test gate. |
| Build | Manual: `npm run build` (CRA + obfuscation) at deploy time. |
| Deployment | Manual `redeploy.sh` : git pull → systemd restart → build → publish static. |
| Rollback | Git revert + re-run deploy (no automated rollback). |
| Migrations | Idempotent, boot-time self-heal + manual scripts under `backend/scripts` . |

> **Recommendation.** Introduce a CI pipeline (lint + test + build on PR) and a CD step (artifact build, blue-green or rolling deploy, automated rollback) — see §53.

| Variable | Purpose |
|---|---|
| NODE_ENV / PORT | Runtime mode; API port (5000). |
| MONGODB_URI | Cosmos vCore connection string (TLS, SCRAM). |
| DB_MAX_POOL | Per-process connection pool cap (default 5). |
| JWT_SECRET / JWT_EXPIRE | Access-token signing & lifetime (30 d). |
| JWT_REFRESH_SECRET / JWT_REFRESH_EXPIRE | Refresh-token signing & lifetime (90 d). |
| COOKIE_EXPIRE | Auth cookie lifetime (days). |
| ENCRYPTION_KEY | AES-256 key (padded/truncated to 32 bytes). |
| CLIENT_URL | Allowed CORS origin (prod frontend). |
| RATE_LIMIT_* / AUTH_RATE_LIMIT_MAX | Rate-limit windows & ceilings. |
| SYNC_INTERVAL_MINUTES | Google sync cadence (15). |
| META_APP_ID / SECRET / VERIFY_TOKEN / SYSTEM_USER_TOKEN | Meta app + webhook + system-user auth. |
| META_API_VERSION / BASE_URL | Graph API version pin (v19.0) & host. |
| META_SYNC_ENABLED / _INTERVAL / _BACKFILL_DAYS | Meta sync toggle, cadence, backfill window. |
| META_BILLING_DRY_RUN | Financial safety switch (default true). |
| GOOGLE_ADS_* (OAuth, dev token, MCC) | Google Ads API credentials. |
| GEMINI_API_KEY / GEMINI_MODEL | AI insights (default gemini-1.5-flash). |
| MAIN_API_URL / META_SYNC_API_URL | Companion-service endpoints (Daily Entry). |
| REACT_APP_API_URL (frontend) | API base URL for the SPA. |

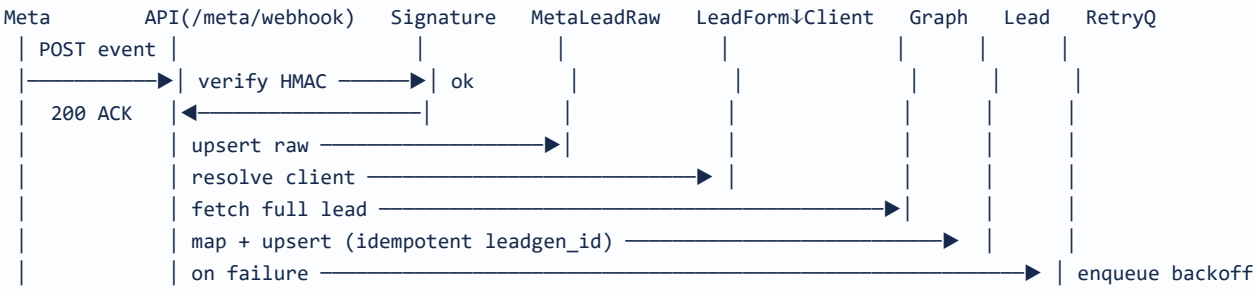
| File | Purpose |
| --- | --- |
| backend/.env(.example) | Backend runtime configuration & secrets. |
| crm-frontend/.env | Frontend build-time config ( `REACT_APP_API_URL` ). |
| backend/package.json | Scripts (start/dev/seed/fix:indexes/oauth), dependencies. |
| crm-frontend/package.json | Scripts (start/build/build:raw/obfuscate/test), deps, browserslist, eslintConfig. |
| backend/config/db.js | Connection, pool sizing, boot self-heal. |
| backend/server.js | Middleware wiring, CORS allow-list, rate limits, route mounting, boot sequence. |
| crm-frontend/scripts/obfuscate.js | Post-build hardening configuration. |
| redeploy.sh | Deployment steps. |
| systemd unit (ara-crm) | Process supervision (host-side). |
| Nginx site config | Static host + reverse proxy + TLS (host-side). |

| Integration | Protocol / auth | Data & direction |
|---|---|---|
| Meta Graph API | REST · system-user token + encrypted page tokens · HMAC webhooks | IN: campaigns, adsets, ads, insights, forms, leads. OUT: page subscribe/unsubscribe. Real-time leadgen webhooks. |
| Google Ads API | gRPC/REST client · OAuth refresh + dev token + MCC | IN: campaigns, daily metrics (incl. impression share, click types), keyword metrics. |
| Google Gemini | REST · API key | OUT: performance snapshot. IN: strict-JSON insight (grade, strengths, strategies). |
| Companion "Main API" & Meta-sync API | REST (Azure-hosted) | Daily-Entry module: clients/leads/funds lookups + trigger meta-sync (cached 5 min). |

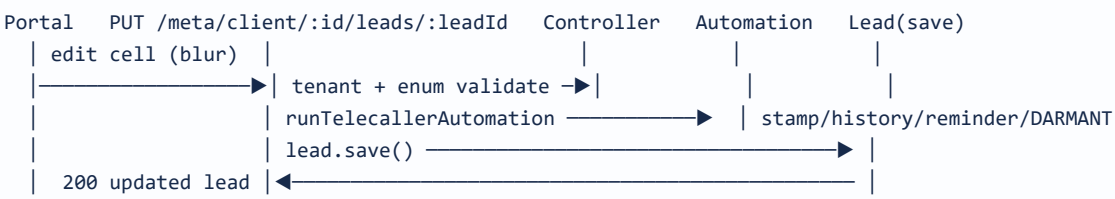
## 40.1 Integration resilience

- Typed errors (auth/permission/rate-limit/transient/validation) drive stop/skip/backoff/retry decisions.
- Exponential backoff (2s/8s/32s) for transient failures; cursor pagination capped (≤10k/edge).
- Rate-limit header parsing (≥80% usage → throttle); sequential per-client processing.
- Webhook retry dead-letter queue + independent poller safety-net.
- AI truncated-JSON repair; graceful 503 when the AI key is absent.

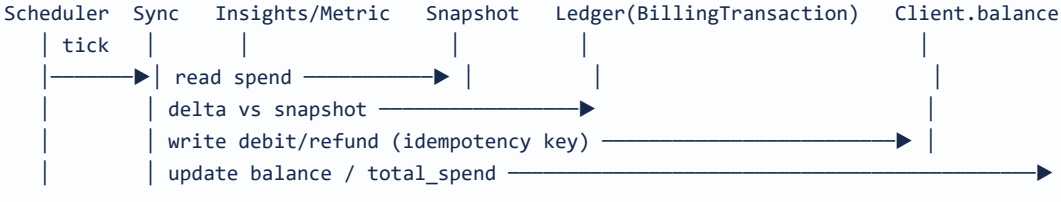
## 41.1 Meta lead ingestion (webhook)



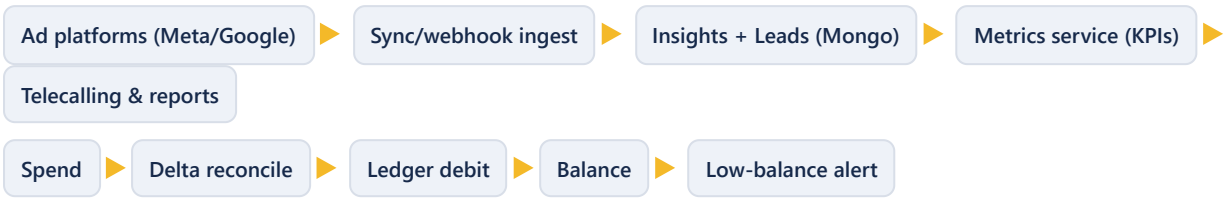

## 41.2 Telecaller save with automation




## 41.3 Billing reconciliation (per campaign/day)



## 42.1 Ad-spend → lead → conversion → billing




## 42.2 Level-0 context


Actors read/write via the SPA → API; the CRM syncs with ad platforms and persists to MongoDB; leads flow in from Meta (push) and both platforms (pull).

## 42.3 Excel round-trip

BACKEND COMPONENTS & DEPENDENCIES

**HTTP SURFACE**

Routers | Middleware (auth/permissions/validation/error)

▼

**CONTROLLERS**

auth | client | lead/telecaller | meta | reports/ai

▼ use

**SERVICES**

metaAds/metaLead/metaBilling | googleAds/analytics | leadXlsx | telecallerAutomation

▼ persist

**MODELS (MONGOOSE)**

24 schemas

▼ trigger

**JOBS**

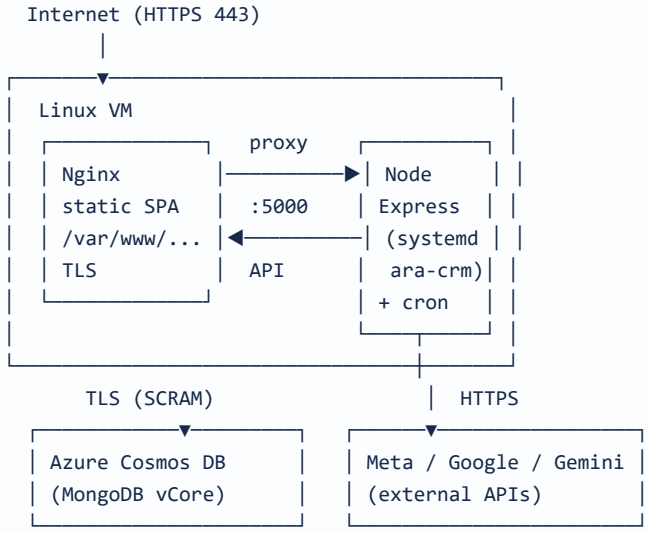
scheduler → syncService / metaSyncService / retry worker


Mongoose "classes" = schemas with instance methods (comparePassword, matchPortalPassword), statics/hooks (pre-save bcrypt/mirror), and virtuals (follow_up_number, reminder_date, month). Sub-documents: follow_ups[], meta_pages[], drop_history[], question_schema[].

```
backend (package)
 ├─ config ──────────▶ models (import for index sync)
 ├─ routes ──────────▶ middleware, controllers
 ├─ controllers ─────▶ services, models, lib, utils
 ├─ services ────────▶ models, utils
 ├─ sync ────────────▶ services, models
 ├─ lib ─────────────▶ models (enums only)
 └─ utils ───────────▶ (leaf: crypto, jwt, mappers)

crm-frontend (package)
 ├─ pages ───────────▶ components, api, store, contexts
 ├─ components ──────▶ api, constants, utils
 ├─ store/slices ────▶ api
 ├─ api ─────────────▶ (axios base: interceptors)
 └─ constants ───────▶ (leaf: enum lockstep w/ backend)
```

Package dependencies are acyclic and point toward leaf utilities/models. The frontend `constants/telecallerSheet` intentionally mirrors backend `models/Lead` enums (kept in lockstep).




| Artifact | Deployed to |
|---|---|
| SPA build (obfuscated) | Nginx web root `/var/www/leadmatrix`. |
| Node API + scheduler | systemd service `ara-crm` :5000. |
| Database | Managed Cosmos DB (external, replicated). |

| Module | Depends on |
|---|---|
| server.js | config/db, all routes, sync/scheduler, middleware. |
| routes/* | middleware (auth/permissions/validation), controllers. |
| controllers/metaController | metaLeadService, metaAdsService, metaBillingService, metaMetricsService, telecallerAutomation, leadXlsxService, Meta* models. |
| sync/scheduler | syncService (Google), metaSyncService, metaLeadService (retry worker). |
| services/metaMetricsService | MetaInsights, Lead (shared KPI authority; consumed by dashboard, leads, analytics). |
| services/*billing* | BillingTransaction, DailyDebitSnapshot, Client, utils. |
| lib/telecallerAutomation | models/Lead enums only (no DB calls — pure functions). |
| utils/* | Leaf modules (crypto, jwt, field mapper, rate-limit, signature). |

**Coupling note.** `metaController.js` is the largest unit (~2,700 LOC) and the highest-fan-out module; a candidate for decomposition into smaller controllers/services (§53).

| Area | Observed convention |
|---|---|
| Language / modules | Modern JS, ES modules (backend `"type":"module"`); async/await throughout. |
| Error propagation | `asyncHandler` wrapper + centralised handler; custom `ErrorResponse`. |
| Naming | camelCase JS; snake_case for ad-platform-sourced fields (mirror external shapes); PascalCase models. |
| Structure | Strict routes→controllers→services→models layering; framework-free domain logic in `lib/`. |
| Documentation | Extensive intent-explaining comments (esp. schemas, billing, automation, self-heal) — a strong maintainability asset. |
| Consistency | Shared enum constants across front/back; single metrics service to prevent KPI drift. |
| Linting | Frontend: CRA ESLint (react-app + jest). Backend: no enforced linter configured. |
| Config | Env-driven with sensible defaults; secrets never committed. |

> **Recommendation.** Add a shared linter/formatter (ESLint + Prettier) and pre-commit hooks across both packages for uniform style enforcement (§53).

| Test type | Current state |
|---|---|
| Unit tests | **Minimal** — only the default CRA `App.test.js` ; no backend unit suite. |
| Integration tests | None automated. |
| Operational smoke tests | Manual scripts under `backend/scripts` (metaSmokeTest, metaSyncSmokeTest, metaWebhookSmokeTest, metaBillingSmokeTest, metaPhase6SmokeTest, testGoogleAds) — run by hand against live integrations. |
| Test tooling present | @testing-library/react + jest-dom (frontend) available but under-used. |

## 49.1 Recommended test pyramid

| Layer | Target |
|---|---|
| Unit | Pure logic first: `telecallerAutomation` (reminder/DARMANT/history), metrics computations, validators, encryption. |
| Integration | API + in-memory Mongo: auth/RBAC, tenant isolation, idempotent ingest, billing reconcile. |
| Contract | Mock Meta/Google/Gemini responses to test error typing & mapping. |
| E2E | Critical journeys (login, connect client, work a lead, view report) via Playwright/Cypress. |

| Ref | Consideration | Status / action |
|---|---|---|
| SC-01 | Vault authorization gap | **Open** — add role/permission gate + access audit to `/api/vault`. |
| SC-02 | Secret management | **Improve** — move keys/tokens from `.env` to a managed secret vault; rotate. |
| SC-03 | Token lifetime & rotation | **Improve** — long-lived access token; wire refresh-exchange & shorten access TTL. |
| SC-04 | Encryption mode | **Note** — AES-256-CBC (no auth tag); consider AES-GCM for integrity. |
| SC-05 | Distributed rate limiting | **Note** — move to shared store before multi-instance. |
| SC-06 | Tenant isolation | **Strong** — token-derived, per-route enforced. |
| SC-07 | Webhook integrity | **Strong** — HMAC verified. |
| SC-08 | Dependency hygiene | **Improve** — add automated dependency/vuln scanning in CI. |
| SC-09 | Audit trail (end-user actions) | **Improve** — add per-field change history beyond financial/lead audit. |

| Ref | Risk | Lik. | Imp. | Mitigation |
|---|---|---|---|---|
| TR-01 | Single-VM SPOF (app + scheduler) | Med | High | Multi-instance API + dedicated worker; systemd restart interim. |
| TR-02 | No CI/CD & minimal tests | High | Med | Introduce CI (lint/test/build) + automated deploy; build test pyramid. |
| TR-03 | API token expiry halts sync/billing | Med | High | Typed errors, alerts, token-refresh runbook, graceful UI. |
| TR-04 | Meta API version deprecation | Med | Med | Centralised version/field config; planned upgrades. |
| TR-05 | Cosmos connection-ceiling exhaustion | Low | High | Small pools; budget pools per instance when scaling. |
| TR-06 | Vault credential exposure | Med | High | Add authorization + auditing (SC-01). |
| TR-07 | metaController monolith (~2.7k LOC) | Med | Low | Refactor into focused controllers/services. |
| TR-08 | In-memory cache/rate-limit under scale-out | Low | Med | Shared Redis cache/limiter store. |
| TR-09 | Manual deploy human error | Med | Med | Automate build/deploy with rollback. |

| Initiative | Benefit |
|---|---|
| Extract job/worker service | Decouple sync/webhook processing from the API; enables API scale-out and isolates load. |
| Message queue for ingestion | Buffer webhook/lead spikes; smooth Graph API pressure; durable retries. |
| Shared cache (Redis) | Consistent caching + distributed rate limiting across instances. |
| Bounded-concurrency sync | Parallelise per-client sync with a concurrency cap as the roster grows. |
| Read replicas / sharding | Scale read-heavy analytics; shard by client for very large datasets. |
| Containerisation + orchestration | Repeatable deploys, autoscaling, rolling updates. |
| CDN for SPA | Global static delivery + edge caching. |

| Ref | Recommendation | Priority | Outcome |
|---|---|---|---|
| RC-01 | Gate & audit the credential vault | P1 | Closes the key authorization gap. |
| RC-02 | Add CI (lint+test+build) & automated CD with rollback | P1 | Safer, repeatable releases. |
| RC-03 | Build the test pyramid (start with pure logic) | P1 | Regression safety on billing & automation. |
| RC-04 | Run ≥2 API instances + extract the scheduler worker | P2 | Removes the SPOF; enables scale-out. |
| RC-05 | Managed secret vault + key rotation + AES-GCM | P2 | Stronger secret handling & integrity. |
| RC-06 | Refresh-token rotation + shorter access TTL | P2 | Reduced token-theft blast radius. |
| RC-07 | Centralised structured logging + monitoring/alerting | P2 | Faster incident detection & resolution. |
| RC-08 | Standardise API error envelope & status codes | P3 | Predictable client integration. |
| RC-09 | Decompose metaController; introduce Redis cache/limiter | P3 | Maintainability & multi-instance readiness. |
| RC-10 | Formalise RTO/RPO, IaC, and a restore drill | P2 | Proven disaster recovery. |

## A · Backend script inventory (operational)

| Category | Scripts |
|---|---|
| Migrations | migrations/2026-04-meta-schema-init, migrateLeadTelecallerFields, fixDailyEntryIndexes, backfillBillingLedger, metaBackfillCreatedTime. |
| Meta onboarding/backfill | bulkFetchAdAccountMeta, bulkFetchPageTokens, bulkSubscribePages, metaAssignPagesToSystemUser, metaPagesManualLink, metaMappingApply/Report, seedMetaClients. |
| OAuth / tokens | getRefreshToken, simpleRefreshToken. |
| Smoke tests | metaSmokeTest, metaSyncSmokeTest, metaWebhookSmokeTest, metaBillingSmokeTest, metaPhase6SmokeTest, testGoogleAds, metaVerifyAccess, inspectMetaActions. |
| Seed | seedAdmin, seedTeams. |

## B · Key npm scripts

| Script | Action |
|---|---|
| backend: start / dev | `node server.js` / `nodemon`. |
| backend: seed:admin / fix:indexes | Seed first admin / repair daily-entry indexes. |
| backend: get-refresh-token / oauth-setup | Google OAuth token acquisition. |
| frontend: build | `react-scripts build` + `obfuscate.js`. |
| frontend: build:raw / start / test | Un-obfuscated build / dev server / CRA tests. |

## C · Technology versions (selected)

| Component | Version |
|---|---|
| Express | 5.x |
| Mongoose | 9.x |
| React / react-router | 19.x / 6.x |
| MUI | 7.x |
| Meta Graph API | v19.0 (pinned) |
| Gemini model | gemini-1.5-flash (default) |

## D · Related documents

- Business Requirements Document — ARA-BRD-CRM-2026-001
- Functional Specification Document — ARA-FSD-CRM-2026-001
- Meta Ads Integration Plan & API Reference; Google Ads Analytics API reference (docs/).

> **End of Document.** This Technical Solution Document describes the ARA Lead Matrix CRM as built and in production as of 17 July 2026. Version 1.0 (baseline); companion to the BRD and FSD; classified Confidential.